
NET 377
Lab 4 – IPSEC Site-Site VPN
PT Answers Document
Due: Sunday, March 1, 2026 by 11:59 pm
50 points total

Your Name: Sukaina Hussain

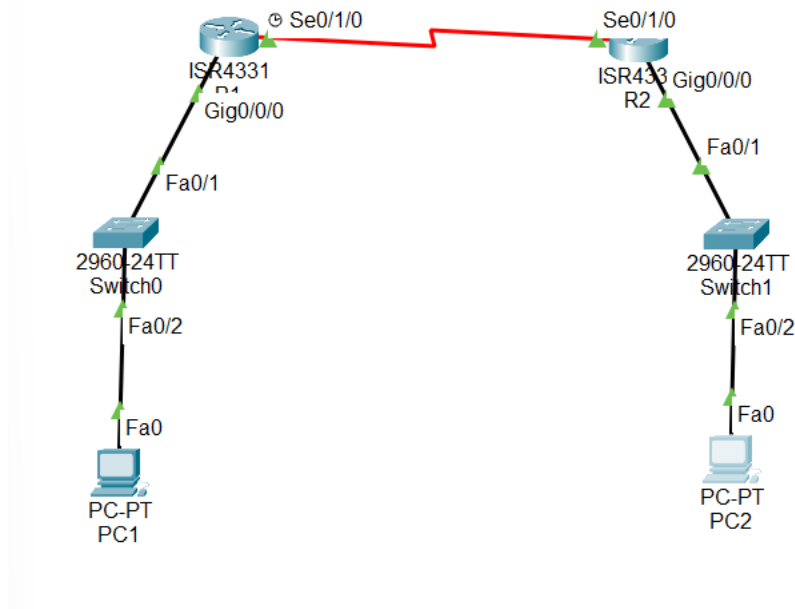
Note: All the requirements for screenshots apply.

All pings should succeed unless otherwise specified.

You can edit this file directly or use a different answer file. The results that you submit on D2L should be in WORD or PDF format.

Fill in All Screenshots and Answers:

1. Paste a screen shot here of your final network with all links green.



2. On R1 and R2, take a screenshot of the commands entered to configure the INBOUND ACL (slide 7).

```
R1-SH#
R1-SH#sh access-lists 110
Extended IP access list 110
  permit udp host 172.30.2.2 host 172.30.2.1 eq isakmp (24 match(es))
  permit esp host 172.30.2.2 host 172.30.2.1 (7 match(es))
  permit icmp any any (4 match(es))
R1-SH#
```

```
R2-SH#sh access-lists 110
Extended IP access list 110
  permit udp host 172.30.2.1 host 172.30.2.2 eq isakmp (49 match(es))
  permit esp host 172.30.2.1 host 172.30.2.2 (7 match(es))
  permit icmp any any (5 match(es))
```

3. On R1 and R2, take a screenshot of the commands entered to configure and show the ISAKMP policy (slide 10).

```
R1-SH#sh crypto isakmp policy

Global IKE policy
Protection suite of priority 10
  encryption algorithm: AES - Advanced Encryption Standard (256 bit keys).
  hash algorithm:      Secure Hash Standard
  authentication method: Pre-Shared Key
  Diffie-Hellman group: #5 (1536 bit)
  lifetime:            86400 seconds, no volume limit
R1-SH#
```

```
R2-SH#sh crypto isakmp policy

Global IKE policy
Protection suite of priority 10
  encryption algorithm: AES - Advanced Encryption Standard (256 bit keys).
  hash algorithm:      Secure Hash Standard
  authentication method: Pre-Shared Key
  Diffie-Hellman group: #5 (1536 bit)
  lifetime:            86400 seconds, no volume limit
R2-SH#
```

4. On R1 and R2, take a screenshot of the commands entered to configure the ISAKMP key (slide 12).

```
R1-SH#show run | include crypto isakmp key
crypto isakmp key NET377-KEY address 172.30.2.2
R1-SH#
```

```
R2-SH#show run | include crypto isakmp key
crypto isakmp key NET377-KEY address 172.30.2.1
R2-SH#
```

5. On R1 and R2, take a screenshot of the commands entered to configure the Interesting Traffic ACL (slide 14).

```
R1-SH#show access-lists 101
Extended IP access list 101
  permit ip 10.0.19.0 0.0.0.255 192.168.19.0 0.0.0.255 (5 match(es))

R2-SH#show access-lists 101
Extended IP access list 101
  permit ip 192.168.19.0 0.0.0.255 10.0.19.0 0.0.0.255
```

-
6. On R1 and R2, take a screenshot of the commands entered to configure the IPSEC transform set (slide 16).

```
R1-SH#show run | section transform-set
crypto ipsec transform-set MYSET esp-aes esp-sha-hmac
set transform-set MYSET
R1-SH#

R2-SH#show run | section transform-set
crypto ipsec transform-set MYSET esp-aes esp-sha-hmac
set transform-set MYSET
R2-SH#
```

7. On R1 and R2, take a screenshot of the commands entered to configure the crypto map (slide 19).

```
R1-SH#show run | section crypto map
crypto map VPNMAP 10 ipsec-isakmp
set peer 172.30.2.2
set transform-set MYSET
match address 101
crypto map VPNMAP
R1-SH#
```

```
R2-SH#show run | section crypto map
crypto map VPNMAP 10 ipsec-isakmp
set peer 172.30.2.1
set transform-set MYSET
match address 101
crypto map VPNMAP
crypto map VPNMAP
R2-SH#
```

8. On R1 and R2, take a screenshot of the commands entered to configure and show the crypto map (slide 21).

```
interface Serial0/1/0
ip address 172.30.2.1 255.255.255.252
ip access-group 110 in
clock rate 64000
crypto map VPNMAP
```

```
interface Serial0/1/0
ip address 172.30.2.2 255.255.255.252
ip access-group 110 in
crypto map VPNMAP
!
```

9. Send interesting traffic:
- On PC1 ping to PC2. Take a screen shot and paste here.

```

C:\>ping 192.168.19.3

Pinging 192.168.19.3 with 32 bytes of data:

Request timed out.
Reply from 192.168.19.3: bytes=32 time=12ms TTL=126
Reply from 192.168.19.3: bytes=32 time=11ms TTL=126
Reply from 192.168.19.3: bytes=32 time=11ms TTL=126

Ping statistics for 192.168.19.3:
    Packets: Sent = 4, Received = 3, Lost = 1 (25% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 11ms, Maximum = 12ms, Average = 11ms

C:\>

```

b. On PC2 ping to PC1. Take a screen shot and paste here.

```

Cisco Packet Tracer PC Command Line 1.0
C:\>ping 10.0.19.3

Pinging 10.0.19.3 with 32 bytes of data:

Reply from 10.0.19.3: bytes=32 time=1ms TTL=126
Reply from 10.0.19.3: bytes=32 time=19ms TTL=126
Reply from 10.0.19.3: bytes=32 time=11ms TTL=126
Reply from 10.0.19.3: bytes=32 time=13ms TTL=126

Ping statistics for 10.0.19.3:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 1ms, Maximum = 19ms, Average = 11ms

C:\>

```

c. On R1 ping to PC2. Take a screen shot and paste here.

d.

```

R1-SH#ping 192.168.19.3

Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 192.168.19.3, timeout is 2 seconds:
!!!!
Success rate is 100 percent (5/5), round-trip min/avg/max = 9/18/51 ms

```

e. On R2 ping to PC1. Take a screen shot and paste here.

```

R2-SH#ping 10.0.19.3

Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 10.0.19.3, timeout is 2 seconds:
!!!!
Success rate is 100 percent (5/5), round-trip min/avg/max = 1/11/43 ms

R2-SH#

```

10. On R1, execute **show crypto isakmp sa**. Take a screen shot and paste here (like slide 23).

```
R1-SH#sh crypto isakmp sa
IPv4 Crypto ISAKMP SA
dst          src          state          conn-id slot status
172.30.2.2    172.30.2.1    QM_IDLE        1085     0 ACTIVE

IPv6 Crypto ISAKMP SA
```

11. On R2, execute **show crypto isakmp sa**. Take a screen shot and paste here (like slide 23).

```
R2-SH#sh crypto isakmp sa
IPv4 Crypto ISAKMP SA
dst          src          state          conn-id slot status
172.30.2.1    172.30.2.2    QM_IDLE        1007     0 ACTIVE

IPv6 Crypto ISAKMP SA

R2-SH#
```

12. On R1, execute **show crypto ipsec sa**. Take a screen shot and paste here (like slide 24).

```
R1-SH#show crypto ipsec sa
interface: Serial0/1/0
  Crypto map tag: VPNMAP, local addr 172.30.2.1

protected vrf: (none)
local  ident (addr/mask/prot/port): (10.0.19.0/255.255.255.0/0/0)
remote ident (addr/mask/prot/port): (192.168.19.0/255.255.255.0/0/0)
current_peer 172.30.2.2 port 500
  PERMIT, flags={origin_is_acl,}
#pkts encaps: 7, #pkts encrypt: 7, #pkts digest: 0
#pkts decaps: 7, #pkts decrypt: 7, #pkts verify: 0
#pkts compressed: 0, #pkts decompressed: 0
#pkts not compressed: 0, #pkts compr. failed: 0
#pkts not decompressed: 0, #pkts decompress failed: 0
#send errors 1, #recv errors 0

local crypto endpt.: 172.30.2.1, remote crypto endpt.:172.30.2.2
path mtu 1500, ip mtu 1500, ip mtu idb Serial0/1/0
current outbound spi: 0xAD9ED2E1(2912867041)

inbound esp sas:
  spi: 0x658463C0(1703175104)
    transform: esp-aes esp-sha-hmac ,
    in use settings ={Tunnel, }
    conn id: 2001, flow_id: FPGA:1, crypto map: VPNMAP
    sa timing: remaining key lifetime (k/sec): (4525504/2035)
    IV size: 16 bytes
    replay detection support: N
    Status: ACTIVE

inbound ah sas:

inbound pcg sas:

outbound esp sas:
  spi: 0xAD9ED2E1(2912867041)
    transform: esp-aes esp-sha-hmac ,
```

-
13. On R2, execute **show crypto ipsec sa**. Take a screen shot and paste here (like slide 24).

```
R2-SH#show crypto ipsec sa

interface: Serial0/1/0
  Crypto map tag: VPNMAP, local addr 172.30.2.2

protected vrf: (none)
local ident (addr/mask/prot/port): (192.168.19.0/255.255.255.0/0/0)
remote ident (addr/mask/prot/port): (10.0.19.0/255.255.255.0/0/0)
current_peer 172.30.2.1 port 500
  PERMIT, flags={origin_is_acl,}
#pkts encaps: 7, #pkts encrypt: 7, #pkts digest: 0
#pkts decaps: 7, #pkts decrypt: 7, #pkts verify: 0
#pkts compressed: 0, #pkts decompressed: 0
#pkts not compressed: 0, #pkts compr. failed: 0
#pkts not decompressed: 0, #pkts decompress failed: 0
#send errors 0, #recv errors 0

local crypto endpt.: 172.30.2.2, remote crypto endpt.:172.30.2.1
path mtu 1500, ip mtu 1500, ip mtu idb Serial0/1/0
current outbound spi: 0x658463C0(1703175104)

inbound esp sas:
  spi: 0xAD9ED2E1(2912867041)
    transform: esp-aes esp-sha-hmac ,
    in use settings ={Tunnel, }
    conn id: 2001, flow_id: FPGA:1, crypto map: VPNMAP
    sa timing: remaining key lifetime (k/sec): (4525504/1968)
    IV size: 16 bytes
    replay detection support: N
    Status: ACTIVE

inbound ah sas:

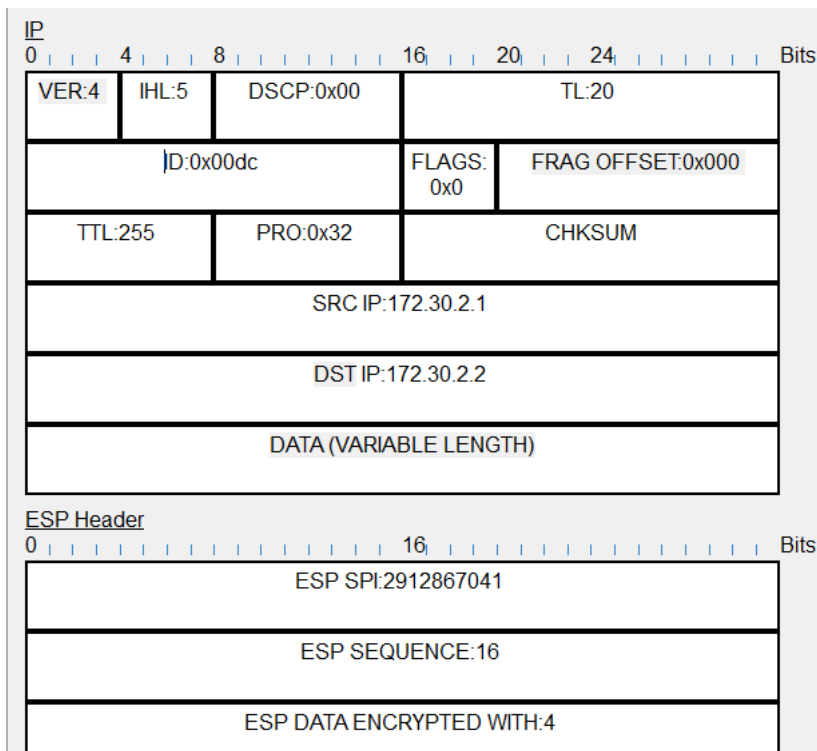
inbound pcsp sas:

outbound esp sas:
  spi: 0x658463C0(1703175104)
    transform: esp-aes esp-sha-hmac ,
    in use settings ={Tunnel, }
```

14. Use Simulation mode to capture an echo request packet flowing across the serial link from R1 to R2. Click on that packet to get the packet details of the IP and ESP headers. Paste in a screen capture here.

a. What is the protocol in the IP header?

ANS: Protocol 50 (0x32)



b. What is the SPI value for the **R1 to R2** direction?

ANS: 2912867041

15. Use Simulation mode to capture an echo reply packet flowing across the serial link from R2 to R1. Click on that packet to get the packet details of the IP and ESP headers. Paste in a screen capture here.

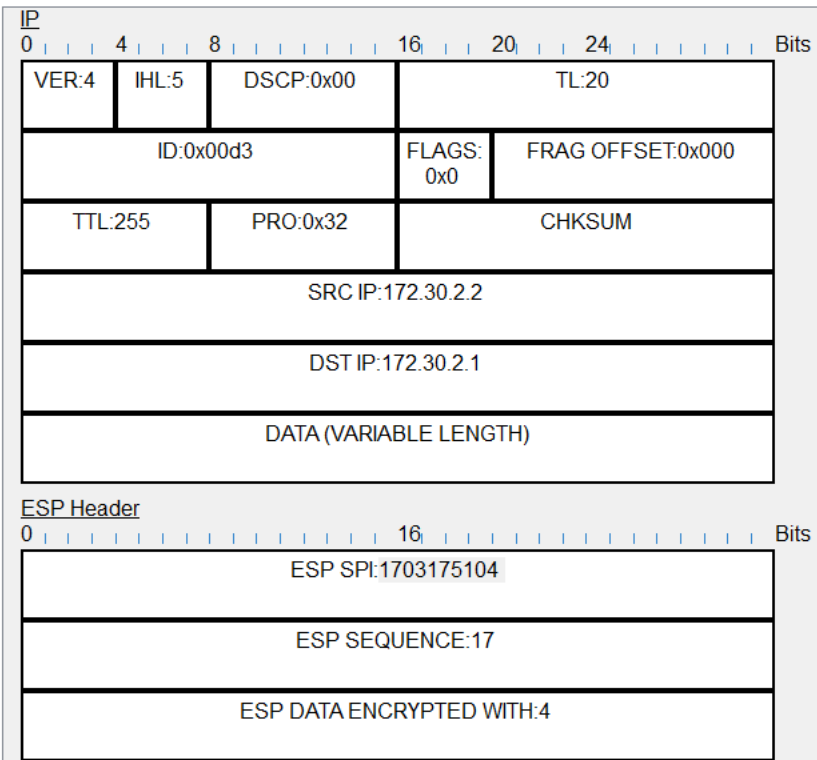
a. What is the protocol in the IP header?

ANS: Protocol 50 (0x32)

b. What is the SPI value for the **R2 to R1** direction?

ANS: 1703175104

PDU Formats



Save a copy of your Answers document in WORD or PDF format. Look through this file to make sure that all screenshot output is clearly readable.

You should submit TWO documents on D2L for Lab 4:

1. Your completed Answers document in WORD or PDF format.
2. Your PKT file.